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Social Demography

Population Studies

Bio-events & Fertility

Fecundity & Fertility

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Fecundity & Fertility

- Fecundity refers to the "capacity of a man, a woman, or a couple to participate in reproduction" - i.e., the capacity to produce a live child.
- It is the physiological capacity to reproduce
- Fertility, on the other hand, refers to "the actual reproductive performance" - whether applied to an individual or a group.
- The fertility of an individual is limited by his/her own fecundity.

Fecundity & Fertility

- The fecundity of an individual may be quite normal, yet the fertility performance may be low,
- Because, fertility can be determined by social, cultural, psychological and even economic factors.
- While a man or a woman or a couple who has given birth to at least one live child is considered fertile
- One who has not had a single child is **sterile.**
- Sterility can be natural/involuntary or artificial/ voluntary.

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Reproductive Span

- A woman becomes biologically fecund with the onset of menstruation.
- The onset of menstruation is called **Menarche**.
- Her capacity to bear children comes to an end with the onset of **Menopause**, when menstruation ceases.
- On the basis of research evidence, women can bear children from age of 15 to 44 years.
- This time is called the reproductive span

Theoretical Maximum Fertility

- Theoretically, during the reproductive span, a woman can bear 37 children, if she gave birth to one child every ten months over a period of 31 years.
- Such a phenomenon is however very rare.
- This reflects a **gap between fecundity and actual fertility.**
- This gap - biological and social factors

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Adolescent Sterility

- The fecundity of females is not uniform throughout the reproductive span of 30 to 35 years.
- Female fecundity is at a low level during the early stages of puberty.
- A certain degree of adolescent sterility (sterility among the adolescent age group) or sub-fertility is observed for girls between the ages of 13 and 18 years.
- Adolescent sterility occurs because regular ovulatory cycles are not generally firmly established for the first two or three years after the onset of menstruation.

Adolescent Sterility

- The interval between menarche and the attainment of full biological maturity to bear children is therefore called the period of adolescent sterility.
- A woman's capacity to bear children reaches the highest level at 20 to 25 years, after which it starts declining first slowly and then after the age of 38 quite rapidly and reaches zero level at about 50 years.

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Post Partum Sterility

- During the reproductive span of women, there are certain periods of temporary sterility.
- After the birth of a child, the woman is generally sterile for some period, as the menstrual cycle is not resumed, or if it is reestablished, the earlier cycles are non-ovulatory.
- This post-partum period in which menstruation is not resumed is known as the **post partum amenorrhea** period.
- During this period, the possibility of conception is very rare and hence this period of temporary sterility is known as the post partum sterility.

Theories of Demography

Biological, Social & Economic

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What is Demography?

- The study of human populations, their growth and decline and various factors like migration, fertility and mortality
- Divided into
  - **FORMAL DEMOGRAPHY** – The Formal Statistical analysis of population parameters and dynamics
  - **POPULATION STUDIES** - Wider investigation of causes and consequences of population structures and change
- Anthropology is interested in Population Studies

Anthropology & Demography

1. Demographic behavior is part of socio-psychological behavior
2. Socioeconomic system guides demographic behavior
3. Problems connected with population are classified "Social Problems"
4. Demography cannot ignore cultural values which provide context in which every population grows and thrives
5. The success and failure of family planning programs are determined by sociocultural consciousness of people

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Malthusian Theory

Assumptions for the theory

- **Food is always necessary for mankind**
  - Population growth is limited by means of subsistence
  - Population increase is proportional to increase in food supply
- **Passion between both the sexes remain at same levels of intensity throughout history**
  - Birth rates remain at relatively high levels
  - Population will proceed in geometric ratios

Malthusian Theory

Relation between two assumptions?

- While food production increases in arithmetic ratio, the population will grow in geometric ratio
- Result –Gap between population and food production continuously widens

Consequences?

- Breakdown of natural ecosystems
- Widening gap between rich and poor

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Malthusian Theory

The Concept of Checks

- **Malthusian Catastrophe** is inevitable (famines and floods) unless CHECKS are present

1. POSITIVE CHECKS

- Largely outside the control of man
- War, disease and poverty

2. PREVENTIVE CHECKS

- Within human control
- Moral restraint, birth control, vice like adultery and pre-marital sex

Malthusian Theory

Critique

1. Mixed purely a biological drive of sex with social instinct of having children
2. Arithmetic and geometric proportions failed empirical validity
3. Too much emphasis on positive checks
4. Unethical preventive checks

Significance

1. Inspired theory building in demography
2. Highlighted the importance of stable land-man ratio

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Theory of Optimum Population Growth

Edwin Canan

1. Population should grow to desired optimum level - Any increase beyond this level is harmful
2. Stage of optimum population is reached when the society is able to run its minimum and essential services
3. This is a stage when Death Rate = Birth Rate
4. There is a close relationship between population size and economic development

Theory of Optimum Population Growth

5. Economic development presupposes optimum utilization of all resources and human resources are employed to exploit natural resources and capital
6. In countries with less population, population growth is important to ensure exploitation of all the resources
7. This situation where all the available resources are fully utilized is OPTIMUM POPULATION
8. In the event of population grows beyond this point, disturbance of natural resources and economic development occurs since more labor are available then what is 'ideally' required.

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Classical Theory – David Ricardo

1. Growing demand for labor is responsible of increase in population
2. As the labor supply increases, wages decrease
3. Ultimately an equilibrium is reached between the demand and supply of labor and everyone gets same wages
4. Beyond this point, if population increases, it leads to universal poverty

Marxism

1. Theory of population is a critique of capitalism
2. The capitalists do not increase their numbers since labor is not an asset to their class
3. Poor accumulate labor because it is their only asset
4. Once the poor uproot the capitalist class, population growth will slow down
5. Communism – A solution to increasing population

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Destiny & Fecundity Principle
Thomas Sadler

The fecundity of human beings is in inverse ratio of the conversion of their numbers

- Two prerequisites for population growth – Labor & Privacy
- With growth and advancement in civilization, the desire to put in labor and attitudes of working hard decrease because of machinery
- Lack of privacy is due to shortage of accommodation
- Hence there is a decrease in fertility

DIET THEORY
THOMAS DOUBLEDAY

- **Man's increase in numbers is inversely proportional to food supply**
- The better the food supply, the slower the increase in population
- Extends his theory to dietary habits
  - Non – Vegetarians – Less capacity to reproduce
  - Vegetarians – Fertility is highest
- Jouse De Castro – "GEOGRAPHY AND HUNGER" drew correlation between protein intake and birth rates of various countries - Poor countries are highly populated

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Demographic Transition

- Considered a more rational approach because it is based on demographic experience of many nations
- Population in any society passes through distinct stages, each with its own peculiarity
- There is however, a divided opinion on number of stages
- Many people contributed to this theory, but the formal form – NOTESTIEN – 1945

Stages of Population Growth

FIRST STAGE - HDR & HBR

High Death Rates

- Poor diets & sanitation
- No medical aid

High Birth Rates

- Illiteracy / Big family advantage
- No family planning
- Early marriages

- Actual growth rate of the population is not very high because high BR is balanced by high DR (high growth potential, though)

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Stages of Population Growth

SECOND STAGE – LDR & HBR

- **Low Death Rate**
  - Rise in income levels and good diets
  - Economic development & Welfare State
  - Good transport etc.
- **High Birth Rate**
  - Accelerated Population growth
  - High growth potential of the first stage is realized here

Stages of Population Growth

THIRD STAGE – LBR & LDR

- **Low Birth Rates and Death Rates**
- Economic development from agrarian to industrial
- Urbanization
- Mobility
- High economic advantage of small families
- Stage of INCIPIENT DECLINE

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Population Dynamics

Fertility, Natality, Mortality

Fertility

Factors & Differentials

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Fecundity, Fertility, Natality

- Fecundity is defined as a "biological potential or the physiological capacity to participate in reproduction."
- Fertility on the other hand refers to the actual reproductive performance of individual or group.
- Natality is the birthrate, which is the ratio of total live births to total population in a particular area over a specified period of time;
- Natality is expressed as childbirths per 1000 people (or population) per year.

Fertility

Factors Determining Fertility

- **Kingsley Davis & Judith Blake – 11 Factors**
- **"Intermediate Variables" –** since any other variable needs to operate "through" them
- Their most basic concept is that the birth of a child is not possible unless...
  1. Sexual intercourse has occurred
  2. Intercourse has resulted in pregnancy
  3. Pregnancy has been brought to successful term.

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Factors in Fertility

1. **Age of Entry into Sexual Unions:**
   - Europe - a couple is not supposed to marry until the husband is able to support a family.
   - Age of marriage - linked to decrease in mortality (man had to wait longer to inherit property) - Ohlin
   - Asia - age at first marriage is very early as the husband is not alone in his responsibilities (extended families)
2. **Permanent Celibacy**
   - A rather high proportion of permanent celibates are frequent in the nations which have a late average age at marriage.

Factors in Fertility

3. **Amount of Reproductive Period Spent in Unions:**
   - Death & divorce rates influence this factor
   - Where monogamy is institutionalized, widows never remarry
   - Further, many divorcees prefer not to remarry or not have children
4. **Voluntary Abstinence**
   - Primitive societies – abstinence during special ceremonies
   - During late pregnancy and early post-partum periods
   - Abstinence during ovulation – "rhythm period"

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Factors in Fertility

5. Involuntary Abstinence

- Men absent themselves from wives in search of gainful labor

6. Frequency of Intercourse

- Responsible for fertility differentials among individuals than population
- Temperature, humidity, chronic diseases, diet etc.

7. Fecundity Due to Involuntary Causes

- STDs – major reason worldwide
- Extreme hunger resulting in amenorrhea in women (temporary sterility) and reduction in sperm count in men

Factors in Fertility

8. Use or Non-use of Contraception

- The most important factor affecting fertility

9. Fecundity due to Voluntary Causes

- Tubectomy & Vasectomy
- Prolonged breast Feeding

10. Fetal Mortality due to Involuntary Causes

- 20% of all pregnancies – spontaneously aborted

11. Fetal Mortality due to Voluntary Causes

- Induced and primitive methods

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Fertility Differentials

- Levels and patterns of fertility vary considerably in various sub-groups of the same population.
- Why study?
  - To identify the factors which determine fertility levels among various sub-groups.
  - To be able to clearly project future population
  - Better targeting of family planning programs
- **Factors**
  - Ecological Factors – Regional & Rural Urban Differences
  - Socioeconomic Factors

Ecological Factors

1. **Regional Differences**
   - Differentials exist among different regions in a nation
   - States and UTs in India for example.
2. **Rural-urban Differentials**
   - Rural areas have more fertility than urban

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Socioeconomic Factors

1. Education attainment, especially of women
2. Economic status of family
3. Occupation of husband
4. Employment of wife
5. Religion, Caste, Race etc.
   - Emphasis on male child
   - Emphasis on procreation
   - Lucknow – Caste study – High Hindus 3.8 and Low 4.1

Mortality

Factors & Differentials

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Mortality

- Study of mortality deals with the effects of death on population
- UN and WHO – ***"Death is permanent disappearance of all evidence of life at any time after birth has taken place"***
- Post Natal cessation of vital functions without capacity of resuscitation
- Historically this factor played a dominant role in determining the size of populations
- More than increase in fertility, decline in mortality is believed to result in high population growth, especially in developed countries

Sex Patterns of Mortality

- Mortality conditions differ for males and females – females more advantageous than the males
- This gap between male and female mortality rates is wide in developed countries
- John Graunt – Father of Demography – 17th Century – ***"Physicians have two women patients to one man, and yet more men die than women"!***
- Is it biological or because of different roles played by them?

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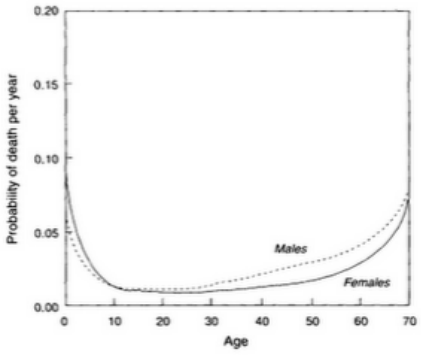
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Sex Patterns of Mortality

- **Conrad and Esterline** conclude that there is little to support a biological explanation and reason for this is purely the roles played by both of them in society
- Man – breadwinner – more physical and emotion stress
- Even retirement has major impact on men than women
- Not much of change in lifestyle even if a woman retires

Age Patterns of Morality

- Typical age specific mortality curve in countries of high mortality is roughly U-shaped
- This indicates mortality is very high at both extremes of life span – infancy and old age

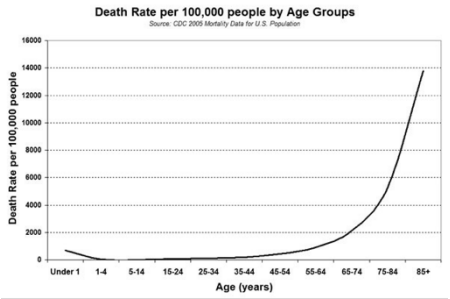


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Age Patterns of Mortality

- In countries of low mortality, the curve is J-shaped due to low IMR



Infant Mortality

- Mortality levels are high in first few hours, days and weeks of life
- Factors affecting fetal and neonatal are primarily endogenous and post neonatal exogenous
- **Endogenous or biological factors**
  - Formation of fetus in womb
  - Age of mother
  - Birth spacing
  - Weight at birth
  - Multiple birth

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Infant Mortality

- **Exogenous Factors**
  - Social, cultural, economic environment
  - Epidemic/communicable diseases
  - Unsanitary conditions
  - Illegitimacy of child – no care
- Developed countries – Endogenous factors while in Developing countries – Exogenous factors

Mortality Differentials

1. Rural and Urban differentials
2. Occupational status
   - Diet
   - Housing
   - Habits
3. Marital status

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Social Consequences of Mortality

1. Low mortality – Reduction in frequency of bereavement – decline in institutions of mourning
2. Decline in mortality – change in the character of religion
3. Decline in mortality – change in family structure – nuclearity
4. High mortality – Reduced interest in interpersonal relations
5. Difference in orientation of time
6. High mortality – high fertility

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